Biochemistry Cume
October 17, 2003

The 2003 Nobel Prize in Chemistry was given for two different but related projects. One of these projects was discussed in one of our Thursday biochemistry seminars in early September.

1. (10 points) Which other Nobel Prize category would have been appropriate for the same research topics?

2. (90 points) Choose one of the two projects that merited the chemistry prize. Discuss why this research was appropriate for a Nobel Prize. In your discussion, include as much detail as time and knowledge permit.

Biochemistry Cumulative Exam

(Supplemental Information Attached)

A mouse is fed a diet including the amino acid aspartate, but no carbohydrate or fatty acids. Small amounts of the essential amino acids and the necessary vitamins and minerals are also included in the diet.

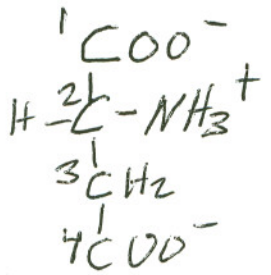
1. (35) Can the mouse synthesize glucose on this diet using aspartate as the sole source of carbon and energy? If the answer is yes, then outline a metabolic pathway for this process. How many aspartate molecules must be used to provide enough carbon atoms, ATP, and NADH molecules to synthesize each molecule of glucose? Write a balanced equation for this process giving the numbers of ATP and CO_2 generated and the number of O_2 consumed.

2. (35) Can the mouse synthesize the fatty acid palmitate, $CH_3(CH_2)_{14}COO^-$, on this diet using aspartate as the sole source of carbon and energy? If the answer is yes, then outline a metabolic pathway for this process. How many aspartate molecules must be used to provide enough carbon atoms, ATP, and NADPH molecules to synthesize each molecule of palmitate? (Assume NADH can be converted into NADPH in the cell.) Write a balanced equation for this process giving the numbers of ATP and CO_2 generated and the number of O_2 consumed.

3. (30) The mouse is fed this diet with aspartate radiolabeled with C-14 at carbon 3. Which carbon atom(s) of glucose and palmitate will first become labeled?

$$CO_2 + NH_4^+ + 3\ ATP + aspartate + 2\ H_2O \longrightarrow urea + 2\ ADP + 2\ P_i + AMP + PP_i + fumarate$$

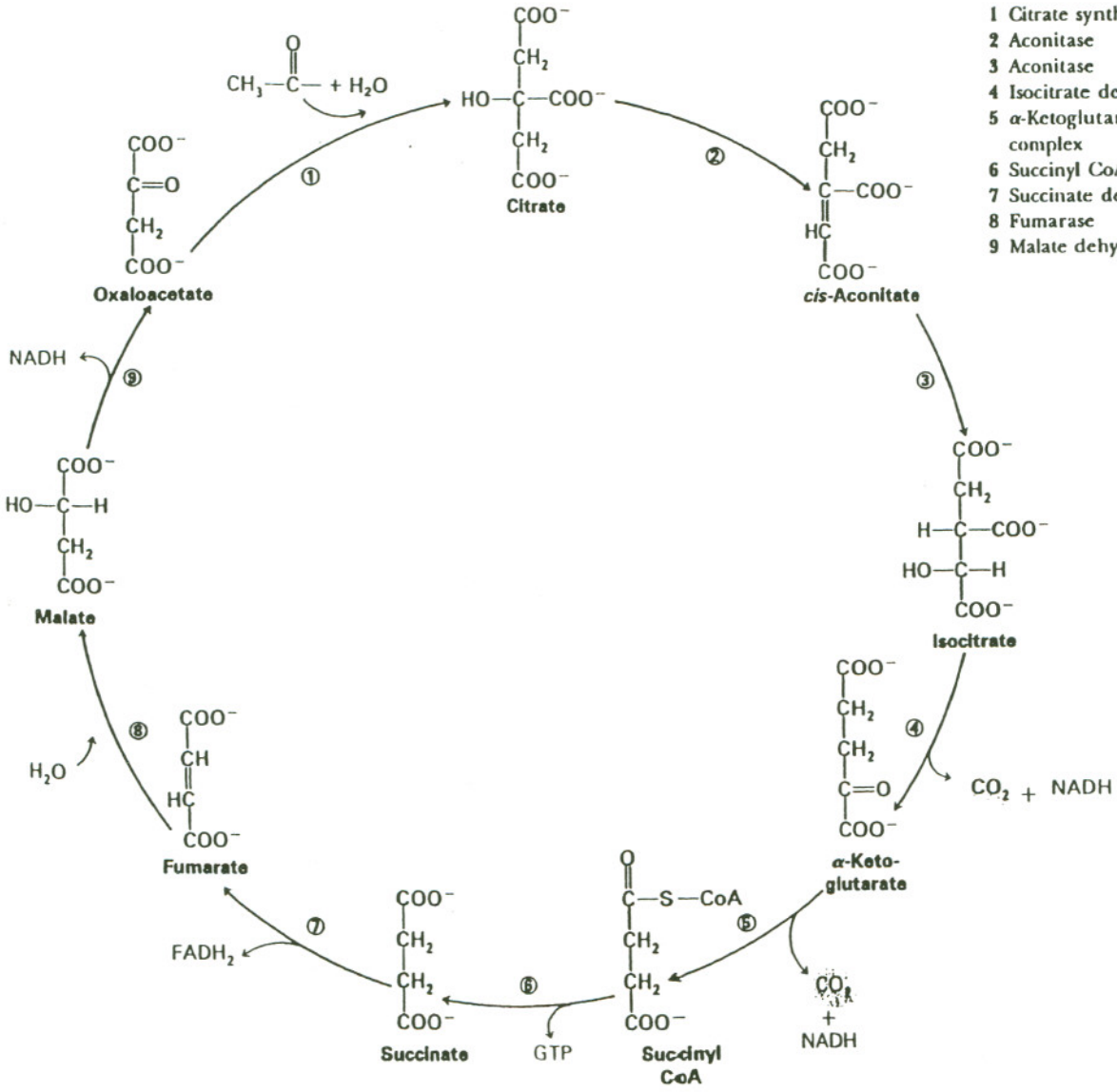
THE UREA CYCLE



Aspartute

following enzymes, as numbered in the diagram:

1 Citrate synthase
2 Aconitase
3 Aconitase
4 Isocitrate dehydrogenase
5 α-Ketoglutarate dehydrogenase complex
6 Succinyl CoA synthetase
7 Succinate dehydrogenase
8 Fumarase
9 Malate dehydrogenase



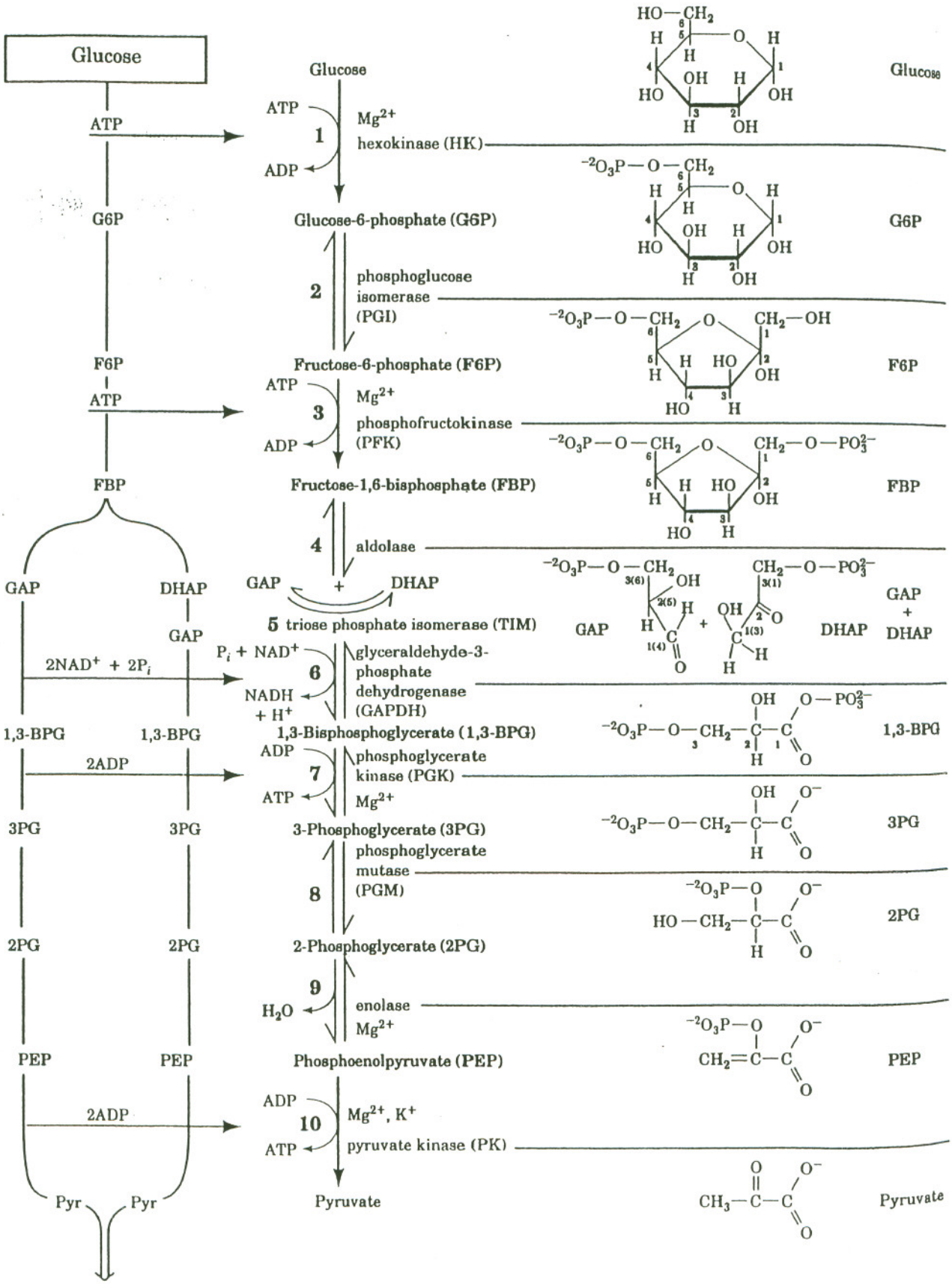

Glucose
ATP
G6P
F6P
ATP
FBP
GAP
DHAP
GAP
2NAD+ + 2Pi
1,3-BPG
1,3-BPG
2ADP
3PG
3PG
2PG
2PG
PEP
PEP
2ADP
Pyr
Pyr
Glucose
ATP
1
ADP
Mg2+
hexokinase (HK)
Glucose-6-phosphate (G6P)
2
phosphoglucose isomerase (PGI)
Fructose-6-phosphate (F6P)
ATP
3
ADP
Mg2+
phosphofructokinase (PFK)
Fructose-1,6-bisphosphate (FBP)
4
aldolase
GAP
+
DHAP
5 triose phosphate isomerase (TIM)
Pi + NAD+
6
NADH + H+
glyceraldehyde-3-phosphate dehydrogenase (GAPDH)
1,3-Bisphosphoglycerate (1,3-BPG)
ADP
7
ATP
Mg2+
phosphoglycerate kinase (PGK)
3-Phosphoglycerate (3PG)
8
phosphoglycerate mutase (PGM)
2-Phosphoglycerate (2PG)
9
H2O
enolase
Mg2+
Phosphoenolpyruvate (PEP)
ADP
10
ATP
Mg2+, K+
pyruvate kinase (PK)
Pyruvate
Glucose
G6P
F6P
FBP
GAP
DHAP
GAP + DHAP
1,3-BPG
3PG
2PG
PEP
Pyruvate

TABLE 22.2 Principal reactions in fatty acid synthesis in bacteria

Step	Reaction	Enzyme
1	Acetyl CoA + HCO_3^- + ATP $\longrightarrow$ malonyl CoA + ADP + P_i + H^+	Acetyl CoA carboxylase
2	Acetyl CoA + ACP $\rightleftharpoons$ acetyl ACP + CoA	Acetyl transacylase
3	Malonyl CoA + ACP $\rightleftharpoons$ malonyl ACP + CoA	Malonyl transacylase
4	Acetyl ACP + malonyl ACP $\longrightarrow$ acetoacetyl ACP + ACP + CO_2	Acyl-malonyl ACP condensing enzyme
5	Acetoacetyl ACP + NADPH + H^+ $\rightleftharpoons$ D-3-hydroxybutyryl ACP + $NADP^+$	β-Ketoacyl ACP reductase
6	D-3-Hydroxybutyryl ACP $\rightleftharpoons$ crotonyl ACP + H_2O	3-Hydroxyacyl ACP dehydratase
7	Crotonyl ACP + NADPH + H^+ $\longrightarrow$ butyryl ACP + $NADP^+$	Enoyl ACP reductase

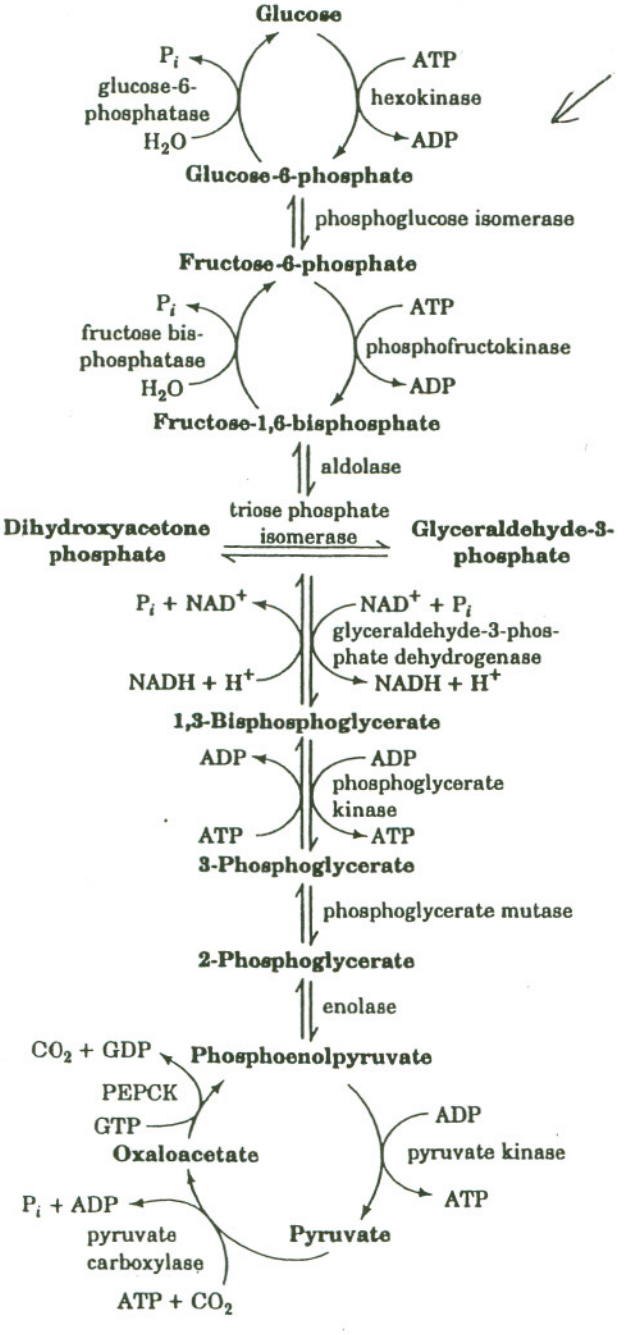


Figure 15-22. Key to Metabolism. **Comparison of the pathways of gluconeogenesis and glycolysis.** The red arrows represent the steps that are catalyzed by different enzymes in gluconeogenesis. The other seven reaction steps of gluconeogenesis are catalyzed by glycolytic enzymes that function near equilibrium.

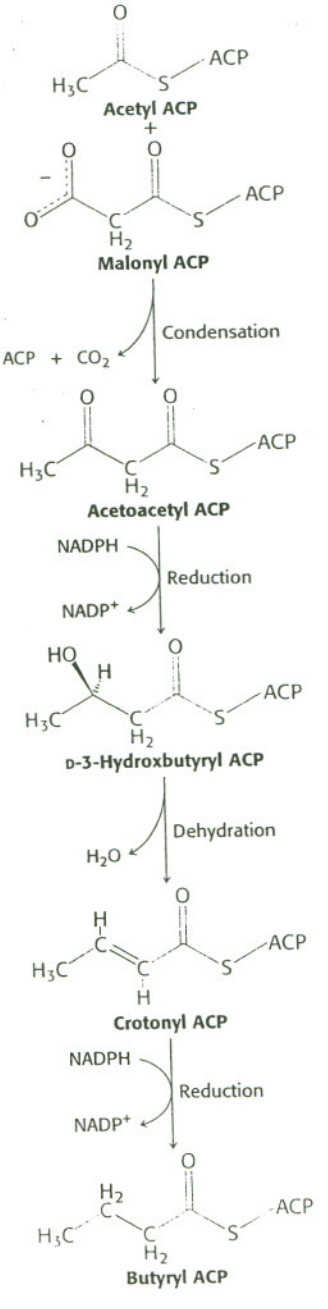


FIGURE 22.22 **Fatty acid synthesis.** Fatty acids are synthesized by the repetition of the following reaction sequence: condensation, reduction, dehydration, and reduction. The intermediates shown here are produced in the first round of synthesis.